

Values-Led Problem Statement

A Sesotho-English Agricultural Advisory Small Language Model for Smallholder Farmers in Lesotho

The Problem

Lesotho's smallholder farmers — who produce most of the country's food on rain-fed, mountainous plots — face planting, irrigation, and pest-management decisions with stakes that are literally life-or-death for their households. Yet the agricultural advice available to them is almost never designed for them:

- **Generic advice, wrong context.** Agricultural information from the internet, radio, and even government extension services is rarely specific to Lesotho's highland climate, its short growing season, or its staple crops (maize, sorghum, beans, and livestock grazing cycles).
- **Language exclusion.** Most advice is in English, while Sesotho is the language of daily life for most farmers — particularly older farmers and women, who make many planting decisions but are least likely to be fluent in English.
- **Poor timing and access.** Advice often arrives too late (after planting windows close), and smartphone-based tools assume cheap data, strong connectivity, and digital literacy that many rural farmers don't have.
- **Distrust of AI output.** Even when farmers can access AI tools, advice that cannot cite a trustworthy local source is often ignored — and rightly so, since a wrong planting recommendation can cost an entire season.

The result: food insecurity, volatile household income, and a widening information gap between farmers with resources (vehicles, smartphones, extension-officer relationships) and those without.

Our Solution

A domain-specific small language model, fine-tuned exclusively on verified agricultural and climate data relevant to Lesotho and Southern Africa, delivered through a lightweight smartphone text app (works offline-first, low-bandwidth) in both Sesotho and English. The model answers practical questions — "When should I plant sorghum in the lowlands this year?" or "What is this pest on my bean leaves?" — with answers grounded in cited, verified sources rather than confident guesses.

Our Guiding Values

Value	What it means in practice
-------	---------------------------

Accessibility	Works on entry-level phones, offline-first, plain-language answers
Linguistic relevance	Sesotho and English parity; agricultural vocabulary validated by native speakers
Accuracy	Grounded in verified sources (Meteorological Services, Ministry of Agriculture, agronomy research); answers cite their basis
Trustworthiness	The model says "I don't know" rather than guessing; advice is checked by extension officers
Community ownership	Farmers and extension officers co-design the data and review outputs — they are stakeholders, not just users

Success Looks Like

- Farmers and extension officers report the advice is usable, timely, and trustworthy.
- Measurable narrowing of the information gap (e.g., % of farmers who can access crop-specific advice in their language before planting).
- The dataset and model are documented transparently so the community, policymakers, and researchers can audit how decisions were made.